

Simon Hathout Willard

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Fourth-year Engineering Physics student at UBC with hands-on experience in industrial automation, embedded systems, and applied AI.

TECHNICAL EXPERIENCE

Research Assistant, UBC Robotics and Control Laboratory | May – August 2026

- Redesigned the AI model-serving infrastructure for the EchoAgent echocardiography agent using NVIDIA Triton, implementing a custom LRU model-loading system to share GPU capacity across multiple vision models and support parallel inference.
- Integrated medical imaging models into an LLM-based agent and developed an interface prototype, allowing visualization of tool outputs and clinician oversight and adjustment.
- Worked with physicians at Vancouver General Hospital to evaluate AI models on point-of-care ultrasound data, supporting benchmarking on a higher-noise clinical dataset.

Automation Engineering Intern at Dometic Vancouver | January – April 2025

- Designed and implemented SCADA software using the Ignition suite, reducing scrap product and improving machine scheduling, leading to savings of \$15000 in 4 months.
- Created and lead training sessions on SCADA software operation, ensuring buy-in from engineering and operations teams.
- Wrote Python and SQL code to process data and create interactive Human Machine Interfaces through Ignition's Vision and Perspective modules.

Automation Subteam Member with Agroponics Design Team | 2023 - 2025

- Used IoT sensor systems with Arduino and ESP32 to measure pH and temperature in a nutrient solution, improving plant growth and productivity.

SKILLS

Programming - Java, Python, SQL, C++, Matlab, Arduino

Software - Git, GitHub, Linux, ROS, Ignition SCADA, NVIDIA Triton

Design Tools - Fusion 360, Onshape, Machining, 3D Printing, Laser Cutting, Woodshop

EDUCATION

4th Year Engineering Physics | 2023 – Present | UBC Applied Sciences

- Courses include quantum physics, thermodynamics, software development, and electro-mechanical design. A full list can be found at <https://hathoutwillard.github.io/courses/>.
- Worked in a team of 4 to design and fabricate a robot with a three-axis arm, to autonomously navigate a course and retrieve objectives.
- Teaching assistant for CPEN 221 (Software Development). Led a lab section, answered discussion board questions, and assisted with grading.
- Cumulative average: 93% (A+)

Exchange Student | Fall 2026 | ETH Zürich Mechanical Engineering

- Courses in nuclear power generation, electric motors, and sustainable power systems.